MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 06 (SURFACE AREAS & VOLUMES)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show clear steps in calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. The volume of a right circular cone whose base area is 156 cm^2 and vertical height is 8 cm is: **[1]**

(a) (a) 2496 cm^3 (b) 1248 cm^3 (c) 1664 cm^3 (d) 416 cm^3

2. Water in a river 3 m deep and 40 m wide is flowing at the rate of 2 km/hr. How much water will fall into the sea in 2 minutes? **[1]**

(a) (a) 800 m^3 (b) 4000 m^3 (c) 8000 m^3 (d) 2000 m^3

3. The volume of the largest right circular cone that can be carved out of a solid cube of edge 2 cm is: **[1]**

(a) (a) $\frac{\pi}{3}$ cm^3 (b) $\frac{8\pi}{3}$ cm^3 (c) $\frac{4\pi}{3}$ cm^3 (d) $\frac{2\pi}{3}$ cm^3

4. The radius of the base and height of a solid right circular cylinder are in the ratio 2 : 3 and its volume is 1617 cm^3. The total surface area of the cylinder is: **[1]**

(a) (a) 770 cm^2 (b) 1155 cm^2 (c) 707 cm^2 (d) 462 cm^2

5. **Assertion (A):** If a right circular cylinder of radius r and height h ($h > 2r$) just encloses a sphere, the diameter of the sphere is $2r$.
Reason (R): The surface area of the enclosed sphere of radius r is $4\pi r^2$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. The volume and total surface area of a solid hemisphere are numerically equal. What is the diameter of the hemisphere? **[2]**

7. A solid metallic sphere of radius 3 cm is melted and recast into the shape of a solid cylinder of radius 2 cm. Find the height of the cylinder. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. An iron pillar has some part in the form of a right circular cylinder and remaining in the form of a right circular cone. The radius of base of each is 8 cm. The cylindrical part is 240 cm high and conical part is 36 cm high. Find the weight of the pillar which weighs 7.8 g per cm^3. **[3]**

9. Find the volume of the largest right circular cone that can be cut out of a solid wooden cube whose edge is 9 cm. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (5 Marks) [5 Marks]

10. Water in a canal, 6 m wide and 1.5 m deep, is flowing with a speed of 10 km/h. How much area will it irrigate in 30 minutes, if 8 cm of standing water is needed for irrigation? **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

11. Case Study: Sports Engineering – Surface Texture of a Dimpled Golf Ball
A tournament golf ball has a diameter of 4.2 cm (radius $R = 2.1$ cm). To reduce aerodynamic drag, its surface is indented with 315 identical hemispherical dimples, each having radius $r = 2$ mm $= 0.2$ cm.

(i) Find the surface area of one hemispherical dimple.
(ii) Find the volume of material carved out to create one dimple.
(iii) Find the total volume of the golf ball before dimples were indented.
(iv) Find the net surface area of the golf ball exposed to the surroundings.

[4]

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